

Environment

Your Android phone is getting better at warning you about earthquakes

An early-warning system on Android phones has detected more than 1,000 earthquakes worldwide since 2021, sometimes giving people 15 to 60 seconds' notice

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By Kasha Patel

Lily Berry was at home alone when her phone began blaring: A 5.1-magnitude earthquake had struck nearby. The August day in 2023 had already been a perilous one, with the remnants of Tropical Storm Hilary passing through the Los Angeles area.

The “hurriquake” felt apocalyptic. When the phone’s high-pitched alert went off, Berry grabbed her dog off the couch and moved to a doorway — and, soon enough, the shaking came.

“Thankfully the magnitude that the alert gave was more or less correct,” said Berry, who has been a resident in the area for more than 30 years. The shaking wasn’t severe at her house, 30 miles from the epicenter in Ojai.

She said the alert “gave me a lot of confidence that I’d likely have time to protect myself and loved ones in case of a future earthquake.”

Berry is one of millions of people who in the past four years have received alerts as part of Android’s native earthquake early-warning system — including during more severe earthquakes greater than a magnitude 7.

A study released Thursday in the journal Science by Google engineers show global Android-based algorithms can detect early seismic activity and provide warnings with similar effectiveness as traditional ground seismic networks, which are not widespread in many earthquake-prone countries.

The system has detected more than 1,000 earthquakes worldwide since 2021, sometimes alerting people 15 seconds to a minute before the shaking propagated toward them.

How it works

The idea emerged from a week-long internal hackathon at Google in 2015. Software engineer Marc Stogaitis laid down the groundwork for a global network using sensors on cellphones, refining it over the years with other engineers.

The system works through a phone's accelerometer, the same feature that twists your phone's screen when you rotate it. That sensor, Stogaitis said, can also detect vibrations — even those originating underground.

Earthquakes generate different types of seismic waves: primary (P) waves that occur first, and secondary (S) waves that are slower and usually more damaging. When an earthquake starts, a phone's sensor can detect the P waves, which many people can't feel. When numerous phones simultaneously detect these vibrations, they send this signal and their location to Google's earthquake detection server. An algorithm assesses whether it's an earthquake, analyzing features such as speed. It also estimates the intensity, which helps the system decide whom to alert before the more noticeable shaking occurs.

The signals from the phone — roughly the speed of light — are traveling much faster than the speed of the earthquake.

The first promising data came in 2018, when the Android system detected a 7.1-magnitude earthquake in Anchorage in 2018. The system began official operations in 2021 and is now available in 98 countries.

 Dive deeper

How to receive alerts

The earthquake early-warning system is on by default on Androids — which make up 70 percent of smartphones around the world — as long as location services are enabled. That means people shouldn't need to do anything more to activate the system. Users can check if the feature is turned on through their phone's settings.

Phones that are stationary and plugged in are best at detecting vibrations from an earthquake.

 Dive deeper

The alert is very hard to miss. The phone emits a loud, high-pitched ring. In the most imminent cases, the alert also includes a message, which can be tested through a phone's settings. These "TakeAction" alerts break through any do-not-disturb settings, take over the entire screen and provide instructions like "Drop, cover and hold on" or "Protect yourself."

Android users farther from the epicenter will probably receive a “BeAware” alert, which appears like a typical Android notification and does not break through the do-not-disturb settings.

The alerts link to a safety page with tips on what to do after an earthquake, as well as more information about the quake provided by Google Search.

It also provides a link to a user survey, which collects information about users’ response to the alert and shaking, detailed in the new study.

How effective are the alerts?

As of March 2024, the system had detected and issued alerts for 1,279 events — and only three were false reports, according to the study. Two were triggered by thunderstorms, but Stogaitis, who is the lead for the earthquake early-warning team, said they have since fixed the issue in the algorithm.

Of the people who received alerts, 85 percent said they experienced shaking. About 36 percent received the alert before ground motion started, 28 percent received it during and another 23 percent received it after.

People typically receive up to 15 seconds of warning for earthquakes estimated with the strongest shaking, of a magnitude 7 or higher. The system gave up to a minute of warning for moderate shaking, of at least magnitude 5.

The system still faces unforeseen blips. In March, an earthquake rocked Myanmar but a service outage in the area before the event prevented signals from being transmitted between cellphones and Google servers, Stogaitis said.

The biggest challenge for the algorithm (and for any earthquake warning system) is estimating the magnitude, especially for stronger earthquakes.

In February 2023, the system initially underestimated the magnitude of an earthquake in Turkey and sent out fewer alerts. Stogaitis said the team has since improved the magnitude accuracy, tracing part of the discrepancy to different types of accelerometers across phones.

“One of the advantages of running a system that is truly global is that we get to record data in many more earthquakes,” said Richard Allen, director of the seismology lab at the University of California at Berkeley and member of the earthquake early-warning team. “We have more large-magnitude earthquakes to continue to learn from.”

The primary goal of the system is to provide early warnings, but Allen said the data is also opening the door for better understanding of the phenomena. The team has seen unbalanced distribution in ground shaking, especially in subduction zones, where one plate slides under another. In the future, they could also create shake maps to better guide emergency response efforts.

“Each time there’s a large earthquake, we try to improve the system and try to learn from it,” Stogaitis said.
